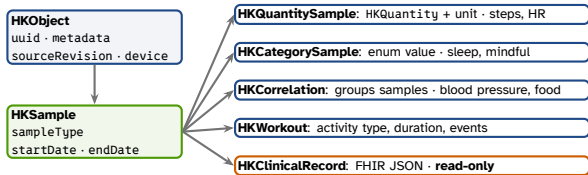


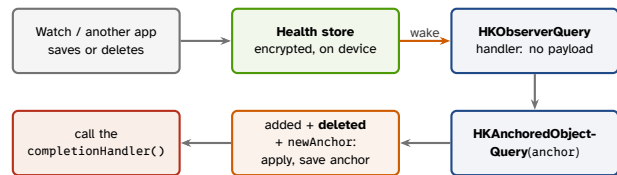
In one line: HealthKit is an **on-device, encrypted store** shared by apps, reached via **one long-lived HKHealthStore**. Access is **per type and per direction** (read / share); **read status is never revealed** — a denied read looks like “no data”. Data = **immutable samples**; totals come from **statistics queries** (they de-duplicate iPhone + Watch), changes from **observer + anchored** ones.

Data model — every object is immutable



Types describe, samples hold values: HKQuantityType(.stepCount) is **cumulative** (sum); heart rate is **discrete** (avg / min / max). **Characteristics** (birth date, blood type) are not samples: read-only, store.dateOfBirthComponents(). Units convert: q.doubleValue(for: .count()), HKUnit(from: "count/min").

Incremental sync — the observer says *that*, the anchor says *what*



Skip the completion → back-off relaunches; **3 misses** → background delivery stops. enableBackgroundDelivery(for:frequency:) (.immediate .hourly .daily .weekly); wakes at most once per period; some types cap at hourly (stepCount on iOS); iOS 15+ needs the com.apple.developer.healthkit.background-delivery entitlement. **Register observers in didFinishLaunching**. A wake while **locked** → reads fail: **retry after unlock**.

How it works

- Setup:** HealthKit capability (+ *Clinical Health Records* only if used); plist NSHealthShareUsageDescription (**read**), NSHealthUpdateUsageDescription (**write**). Xcode adds healthkit to *required device capabilities* — remove it if optional.
- Availability:** isHealthDataAvailable() first. iPhone, Watch, visionOS; iPad only from **iPadOS 17**; macOS links the framework but gets false. **One store per app**, kept.
- Authorization:** requestAuthorization(toShare:read:), per type. The sheet shows only for types *not yet decided*, then returns at once. **Success = processed, not granted**. authorizationStatus(for:) reports **share only**; getRequestStatusForAuthorization says whether a prompt would show.
- Read denied** ⇒ you see only *your own* samples — so “has diabetes data” cannot leak through an error.
- Writing:** build an HKQuantitySample, save. You may **delete only what you saved**. To “edit”: HKMetadataKeySyncIdentifier + a higher HKMetadataKeySyncVersion → HealthKit replaces the old sample. HKMetadataKeyWasUserEntered marks manual entries.
- Locked device:** reads fail with errorDatabaseInaccessible; **saves still work** — staged, merged on unlock.

Example — steps per day, last 7 days (async, iOS 15.4+)

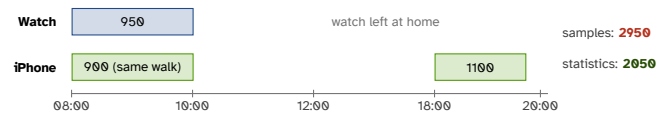
```

let store = HKHealthStore() // one per app, keep it
let steps = HKQuantityType(.stepCount)
guard HKHealthStore.isHealthDataAvailable() else { return }
try await store.requestAuthorization(toShare: [], read: [steps])
// returning = sheet dismissed, NOT "granted"; denied read = empty
let cal = Calendar.current
let start = cal.date(byAdding: .day, value: -6, to: cal.startOfDay(for: .now))!
let week = HKQuery.predicateForSamples(withStart: start, end: .now)
let q = HKStatisticsCollectionQueryDescriptor(
  predicate: .quantitySample(type: steps, predicate: week),
  options: .cumulativeSum, // merges iPhone + Watch overlap
  anchorDate: start, // buckets start at midnight
  intervalComponents: DateComponents(day: 1))
let days = try await q.result(for: store) // results(for:) = live updates
days.enumerateStatistics(from: start, to: .now) { s, _ in
  print(s.startDate, s.sumQuantity()?.doubleValue(for: .count()) ?? 0) }
  
```

Workouts

watchOS: HKWorkoutSession + HKLiveWorkoutBuilder. **iOS 17:** HKWorkoutSession on iPhone/iPad (a Watch session can startMirroringToCompanionDevice). **iOS 26:** HKLiveWorkoutBuilder on iPhone/iPad — standalone: prepare() (countdown) → startActivity + beginCollection → delegate streams metrics → stopActivity, end, finishWorkout(). Heart rate needs an external strap; the phone locks → show a Live Activity. Logging one after the fact: HKWorkoutBuilder.

Why summing raw samples over-counts



Statistics queries merge overlapping sources, as Health does; summing raw samples (or .separateBySource totals) double-counts.

The query family

HK...Query	Returns	Use for
Sample	snapshot; sort, limit	latest readings
Statistics	one sum / avg / min / max	today's total
StatisticsCollection	a statistic per interval	steps per day
AnchoredObject	adds + deletes + anchor	server sync
Observer	"changed" — no data	trigger, wake

Async ...QueryDescriptor (iOS 15.4+): result(for:) one-shot, results(for:) an AsyncSequence of updates. Persist HKQueryAnchor (NSSecureCoding).

Clinical records and App Review

HKClinicalRecord: FHIR (allergies, meds, labs), **read-only**, own plist key. **5.1.3(i):** no health data for ads, marketing or data mining; third parties only for health research, with permission. **(ii):** no false data, **no personal health information in iCloud**. **2.5.1:** health/fitness use, integrate with Health.

Traps

- ! “Was read denied?” is unknowable — design the empty state instead.
- ! Observer created in a view controller: a background wake finds no query. Register at launch.
- ! doubleValue(for:) with an incompatible unit throws an ObjC exception — a crash, not a Swift error.
- ! Buckets anchored at “now” split days at 14:32 — anchor at midnight.

Remember

Write status is visible, read status never is; statistics merge, samples don't; the observer says *that*, the anchor says *what*.